Food Remedy API

RE001 – Competitive App Research: Grocery Coach

Prepared For: Food Remedy API Team

Author: Riken patel

Date: 14 August 2025 (T2 2025)

Target Audience: Users with food allergies or dietary intolerances (e.g., gluten, dairy, nuts, soy)

1. Objective of Research:

- Evaluate how the app scans groceries for allergens
- Measure barcode recognition speed
- Assess the user interface (UI) layout and usability
- Analyze the presentation of allergy warnings and food safety information
- Identify unique features that can inspire new app functionality

2. Barcode Scanning Performance

Observations:

- **Recognition Speed:** The barcode scanner identifies most products in approximately 1–2 seconds under normal lighting conditions.
- **Accuracy:** High accuracy for commonly available supermarket items. In testing, 8 out of 10 scanned products were correctly recognized.
- **Scan Range:** Effective from 10–20 cm distance, autofocus works reliably.
- **Fallback Option:** If a barcode is unrecognized, the app allows manual product search by name or brand.

Conclusion:
The scanning function is fast and reliable, providing a seamless user experience for identifying grocery items quickly.

3. **Allergen Detection**

Features:

- Users can set up an **allergen profile** (e.g., gluten, nuts, dairy, soy) during onboarding or in profile settings.
- Scanned products are analyzed for these allergens. If present, they are highlighted in **red** and labeled clearly (e.g., “Contains gluten”).
- The app also warns of hidden allergens and cross-contamination (e.g., “May contain traces of nuts”).
- Food certifications such as **gluten-free**, **vegan**, or **dairy-free** are displayed where applicable.

Effectiveness:

- The allergen detection is thorough and helps identify uncommon ingredient names that may cause allergic reactions.
- Alerts are visually clear and easy to understand.

4. **User Interface (UI) and User Experience (UX)**

Observations:

- **Design:** Clean, modern interface with soft colors and intuitive icons.
- **Navigation:** Bottom tab bar includes Scan, Search, My Preferences, and Saved Items.
- **Accessibility:** High-contrast buttons and readable fonts suitable for all age groups.
- **Onboarding:** Step-by-step guidance and tooltips make the app easy for first-time users.

Conclusion:
The app’s UI is user-friendly, intuitive, and well-organized, which supports quick allergen identification.

5. **Allergy Warnings and Food Safety Information**

Observations:

- Detected allergens are highlighted in red with an **exclamation mark icon** for easy recognition.
- Safe items are marked with a green check.

- Each product detail page shows:
  - Full ingredient list
  - Manufacturer information
  - Processing warnings (e.g., shared facilities)
  - User-submitted experiences regarding allergen reactions

Conclusion:
The app communicates allergy and food safety information clearly, helping users make informed decisions.

6. **Unique Feature: Crowd-Sourced Allergy Feedback**

Description:

- Users can rate scanned products based on their personal experience with allergens:
  - ✅ No reaction
  - ⚠️ Mild reaction
  - ❌ Severe reaction
- This generates a **community-based allergen safety score** for each product.

Potential Inspiration for Our App:

- A similar **community trust or safety rating** system could be implemented to allow users to share real-world allergy experiences, adding an additional layer of safety beyond just label information.

7. **Summary of Key Findings**

Feature	Observation	Potential Application in Our App
Barcode Scanning	Fast and accurate	Implement optimized scanning with manual fallback
Allergen Detection	Personalized allergen profile and clear alerts	Include color-coded allergen warnings
UI/UX	Clean, intuitive layout with easy navigation	Design similar user-friendly interface

Allergy Warnings & Food Info	Visual cues, ingredient list, certifications, user feedback	Display clear warnings and add community experience section
Unique Feature	Crowd-sourced allergen feedback	Community trust rating for allergen safety

8. Conclusion

The *Grocery Coach* app provides a comprehensive solution for users with allergies by combining fast barcode scanning, accurate allergen detection, clear UI, and crowd-sourced safety feedback. Its approach to community-based allergen ratings is particularly innovative and could be directly applied to enhance the functionality of our own grocery safety app.